File

Name: ex7-diag-lim
Description: TR30 example 7 ex7-diag-lim - UNACCEPTED
Comment: DIAGNOSTIC ONLY: printed TR30 strength limits used explicitly to isolate material-limit differences.
Changed by: AAA on: 29.10.2026 at: 15:49:32

1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1 ----------	Gear 2 -------
Power (kW)	[P]	298.577	
Speed (1/min)	[n]	5400.0	1082.5
Number of load cycles (in mio.)	[NL]	9720.000	1948.531
Torque (Nm)	[T]	528.0	2633.9
Application factor	[KA]	1.00	
Required service life (h)	[H]	30000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1 ----------	Gear 2 -------
Center distance (mm)	[a]	460.375	
Center distance allowances (mm)	[Aa.e/i]	0.000 / 0.000	
Normal module (mm)	[mn]	1.6933	
Transverse module (mm)	[mt]	1.7836	
Normal diametral pitch (1/in)	[Pnd]	15.00000	
Transverse diametral pitch (1/in)	[Ptd]	14.24056	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	18.3100	
Number of teeth	[z]	86	429
Double helical gearing		left/right	right/left
Total facewidth of Gear (mm)	[B]	215.90	215.90
Width of intermediate groove (mm)	[bNut]	50.80	50.80
Facewidth for calculation (mm)	[beH]	82.55	82.55
Accuracy grade	[Q-ISO 1328:2013]	A5	A5
Inner diameter (mm)	[di]	0.00	0.00
Shaft diameter (mm)	[ds]	77.58	377.50
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1 ----------	Gear 2 -------

3 Materials

Gear 1

Own input, TR30 example 7 V alloy steel ML, Through hardened steel, alloyed/through hardened, Original TR30 table properties; material limits from ISO 6336-5 Table 1

Gear 2

Own input, TR30 example 7 V alloy steel ML, Through hardened steel, alloyed/through hardened, Original TR30 table properties; material limits from ISO 6336-5 Table 1

		------- Gear 1	---------- Gear 2 -------
Surface hardness		HBW 342	HBW 300
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	256.00	238.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	661.00	603.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	1000.00	1000.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	1.00	1.00
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	6.00	6.00
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	18.00	18.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Oil type, own input	TR30 example 7: nu40 input; other oil properties are ancillary defaults	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	25.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	19.00
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	80.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile, own input	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472
Addendum coefficient	[haP*]	1.001
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	77.309
(mm)	[a0.e/i]	77.011 /77.011
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.661 / 0.661

Reference profile Gear 2

Reference profile, own input	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.472

Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	383.075
(mm)	[a0.e/i]	382.777 /382.777
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.648 / 0.648

4.1.1 Information on final machining

		------- Gear 1 ----------	------- Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.001	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000

Type of profile modification:	none (only running-in)		
Tip relief by running in (µm)	[Ca L/R]	13.0 /13.0	13.0 /13.0

4.2 Basic data

Overall transmission ratio	[itot]	-4.988
Gear ratio	[u]	4.988
Transverse module (mm)	[mt]	1.784
Transverse pressure angle (°)	[αt]	20.976
Working pressure angle (°)	[αwt]	21.326
(°)	[αwt.e/i]	21.326 /21.326
Working pressure angle at normal section (°)	[αwn]	20.333
Base helix angle (°)	[βb]	17.170
Helix angle at operating pitch circle (°)	[βw]	18.350
Reference center distance (mm)	[ad]	459.287
Sum of profile shift coefficients	[Σxi]	0.6478

		------- Gear 1 ----------	------- Gear 2 -------
Profile shift coefficient	[x]	0.3615	0.2863
Generating profile shift coefficient	[xE.e/i]	0.1857/ 0.1857	0.1104/ 0.1104
Virtual gear no. of teeth	[zn]	99.235	495.019
Involute length (mm)	[l_dFa-l_dFf]	3.593	3.609

4.3 Diameters and their allowances

		------- Gear 1 ----------	------- Gear 2 -------
Reference diameter (mm)	[d]	153.393	765.181
Base diameter (mm)	[db]	143.228	714.473
Tip alteration (mm)	[k*mn]	-0.009	-0.009
Tip diameter (mm)	[da]	157.990	769.520
(mm)	[da.e/i]	157.990 /157.990	769.520 /769.520
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / 0.000	0.000 / 0.000
Tip form diameter (mm)	[dFa]	157.990	769.520
(mm)	[dFa.e/i]	157.990 /157.990	769.520 /769.520

Active tip diameter (mm)	[dNa]	157.990	769.520
(mm)	[dNa.e/i]	157.990 /157.990	769.520 /769.520
V-Circle diameter (mm)	[dv]	154.617	766.150
(mm)	[dv.e/i]	154.022 /154.022	765.554 /765.554
Operating pitch diameter (mm)	[dw]	153.756	766.994
(mm)	[dw.e/i]	153.756 /153.756	766.994 /766.994
Root diameter (mm)	[df]	150.384	761.917
(mm)	[df.e/i]	149.788 /149.788	761.321 /761.321
Active root diameter (mm)	[dNf]	151.392	763.144
(mm)	[dNf.e/i]	151.392 /151.392	763.144 /763.144
Root form diameter (mm)	[dFf]	151.336	762.790
(mm)	[dFf.e/i]	150.807 /150.807	762.208 /762.208

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.422	0.423
Effective tip clearance (mm)	[c.e/i]	0.719 / 0.719	0.721 / 0.721
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.293 / 0.293	0.468 / 0.468
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	2.299	2.170
(mm)	[ha.e/i]	2.299 / 2.299	2.170 / 2.170
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	1.505	1.632
(mm)	[hf.e/i]	1.802 / 1.802	1.930 / 1.930
Tooth height (mm)	[h]	3.803	3.802

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	26.676 /26.676	22.920 /22.920
Roll angle to dNf (°)	[ξNf.e/i]	19.620 /19.620	21.506 /21.506
Roll angle at dFf (°)	[ξFf.e/i]	18.884 /18.884	21.291 /21.291

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8339	1.7792
Normal tooth thickness at tip circle (mm)	[san]	1.310	1.411
(mm)	[san.e/i]	1.088 / 1.088	1.193 / 1.193
Normal tooth thickness at tip form circle (mm)	[sFan]	1.310	1.411
(mm)	[sFan.e/i]	1.088 / 1.088	1.193 / 1.193
Normal space width at root circle (mm)	[efn]	1.176	1.132
(mm)	[efn.e/i]	1.207 / 1.207	1.138 / 1.138
Pitch on reference circle (mm)	[pt]	5.603	5.603
Base pitch (mm)	[pbt]	5.232	5.232
Transverse pitch on contact-path (mm)	[pet]	5.232	5.232
Lead height (mm)	[pz]	1456.272	7264.425
Axial pitch (mm)	[px]	16.933	16.933

4.7 Sliding

		------- Gear 1	---------- Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	3.654	2.333
Specific sliding at the tip	[ζa]	0.194	0.144
Specific sliding at the root	[ζf]	-0.168	-0.240
Mean specific sliding	[ζm]	0.174	
Sliding factor on tip	[Kga]	0.084	0.054
Sliding factor on root	[Kgf]	-0.054	-0.084

4.8 Contact ratios

		---------- Pair ----------
Minimal length of contact line (mm)	[Lmin]	144.941
Transverse contact ratio	[εα]	1.686
	[εα.e/m/i]	1.686 / 1.686/ 1.686
Overlap ratio	[εβ]	4.875
Total contact ratio	[εγ]	6.561
	[εγ.e/m/i]	6.561 / 6.561/ 6.561
Length of path of contact (mm)	[gα]	8.819
(mm)	[gα.e/m/i]	8.819 / 8.819/ 8.819
Approach path of contact (mm)	[gf]	3.437
(mm)	[gf.e/m/i]	3.437 / 3.437/ 3.437
Recess path of contact (mm)	[ga]	5.383
(mm)	[ga.e/m/i]	5.383 / 5.383/ 5.383

		------- Gear 1	---------- Gear 2 -------
Addendum contact ratio	[ε]	1.029	0.657
	[ε.e/i]	1.029 / 1.029	0.657 / 0.657
Length T1-A and T2-A (mm)	[T1A,T2A]	24.522	142.907
(mm)	[.e/i]	24.522 / 24.522	142.907 /142.907
Length T1-B and T2-B (mm)	[T1B,T2B]	28.110	139.319
(mm)	[.e/i]	28.110 / 28.110	139.319 /139.319
Length T1-C and T2-C (mm)	[T1C,T2C]	27.959	139.470
(mm)	[.e/i]	27.959 / 27.959	139.470 /139.470
Length T1-D and T2-D (mm)	[T1D,T2D]	29.754	137.674
(mm)	[.e/i]	29.754 / 29.754	137.674 /137.674
Length T1-E and T2-E (mm)	[T1E,T2E]	33.342	134.087
(mm)	[.e/i]	33.342 / 33.342	134.087 /134.087
Length T1-T2 (mm)	[T1T2]	167.429	
(mm)	[.e/i]	167.429 /167.429	
Diameter of single contact point B (mm)	[d-B]	153.866	766.884
(mm)	[d-B.e/i]	153.866 /153.866	766.884 /766.884
Diameter of single contact point D (mm)	[d-D]	155.098	765.695
(mm)	[d-D.e/i]	155.098 /155.098	765.695 /765.695

5 General influence factors

5.1 Forces and circumferential speed

		------- Gear 1	---------- Gear 2 -------
Nominal circum. force at pitch circle (N)	[Ft]	6884.28	
Axial force (N)	[Fa]	0.00	0.00
Radial force (N)	[Fr]	2639.30	2639.30
Normal force (N)	[Fnorm]	7716.80	7716.80

Nominal circumferential force per mm (N/mm)	[w]	41.70	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	6868.01	
Axial force (N)	[Faw]	0.00	0.00
Radial force (N)	[Frw]	2681.36	
Circumferential speed reference circle (m/s)	[v]	43.37	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	43.47	

5.2 Contact stiffness

Running-in value (µm)	[yp]	1.5
Running-in value (µm)	[yf]	1.5
Correction factor	[CM]	0.800
Gear blank factor	[CR]	1.000
Basic rack factor	[CB]	0.975
Material coefficient	[E/Est]	1.000
Single stiffness (N/mm/µm)	[c']	12.730
Theoretical single stiffness (N/mm/µm)	[c'th]	21.392
Contact stiffness (N/mm/µm)	[cγα]	19.275
Contact stiffness (N/mm/µm)	[cγβ]	16.384
Reduced mass (kg/mm)	[mRed]	0.08138
Resonance speed (min-1)	[nE1]	1709
Resonance ratio (-)	[N]	3.160
Overcritical range		

5.3 Calculation of K factors

Running-in value (µm)	[yα]	1.5	
Bearing distance l of pinion shaft (mm)	[l]	355.600	
Distance s of pinion shaft (mm)	[s]	0.000	
Outside diameter of pinion shaft (mm)	[dsh]	63.500	
Load in accordance with Figure 13, ISO 6336-1:2006 0:a), 1:b), 2:c), 3:d), 4:e)	[-]		0
Coefficient K' according to Figure 13, ISO 6336-1:2006	[K']		0.48
With stiffening			
Effective flank line deviation (µm)	[Fβy]	3.27	
from deformation of shaft (µm)	[fsh·B1]	1.40	
fsh (µm)	[fsh]	1.40	
B1	[B1]	1.00	
fHβ5 (µm)	[fHβ5]	8.50	
Tooth without flank line modification			
Position of contact pattern:	favorable		
from production tolerances (µm)	[fmα*B2]	11.67	
B_2=1.00			
Tooth trace deviation, theoretical (µm)	[Fβx]	6.63	
Running-in value (µm)	[yβ]	3.37	

5.4 K factors

Dynamic factor	[Kv]	1.684
Face load factors		
- Flank	[KHβ]	1.381
- Tooth root	[KFβ]	1.361
- Scuffing	[KBβ]	1.381
Transverse load factors		
- Flank	[KHα]	1.369

- Tooth root	[KFα]	1.369	
- Scuffing	[KBα]	1.369	
Application factor	[KA]	1.000	
Mesh load factor	[Kγ]	1.000	

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.1857	0.1104
Tooth form factor	[YF]	0.857	0.894
Stress correction factor	[YS]	2.297	2.358
Load application angle (°)	[αFen]	20.410	20.099
Load distribution influence factor	[fε]	0.736	
Load application diameter, virtual spur gear (mm)	[d_{en}]	169.391	839.605
Load application diameter (mm)	[d_{en}]	154.746	766.552
Bending moment arm (mm)	[hFe]	1.720	1.874
(-)	[hFe*]	1.016	1.107
Tooth root thickness at critical cross section (mm)	[sFn]	3.869	3.958
(-)	[sFn*]	2.285	2.338
Root fillet radius at critical cross section (mm)	[ρF]	0.738	0.673
(-)	[ρF*]	0.436	0.397
Tangent contact point, virtual spur gear (mm)	[x,y]	1.934 /75.198	
Tangent contact point, virtual spur gear (mm)	[x,y]		1.979 /380.980
Diameter at critical cross section (mm)	[d_{sFn}]	150.445	761.971
Tangent at critical cross section (°)	[$α_{sFn}$]	30.000	30.000
Notch parameter	[q_s]	2.620	2.941
Helix angle factor	[Yβ]	0.990	
Deep tooth factor	[YDT]	1.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	82.55	82.55
Nominal stress at tooth root (N/mm²)	[σF0]	48.02	51.38
Tooth root stress (N/mm²)	[σF]	150.63	161.18
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	1.001	1.003
Surface factor	[YRrelT]	0.964	0.964
Size factor, tooth root	[YX]	1.000	1.000
Life factor	[YNT]	0.850	0.878
$Y_{δrelT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.820	0.849
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	2.00	
$Y_{ST} \cdot σ_{Flim}$ (N/mm²)	[σFE]	512.00	476.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	313.78	301.87
Limit strength tooth root (N/mm²)	[σFG]	420.05	404.10

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.34	1.34
Safety for tooth root stress	[SF=σFG/σF]	2.79	2.51
Transmittable power (kW)	[kWRating]	621.99	559.19

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------

Zone factor	[ZH]	2.369	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.770	
Helix angle factor	[Zβ]	1.026	
Effective facewidth (mm)	[beff]	82.55	
Nominal contact stress (N/mm²)	[σH0]	203.09	
Contact stress at operating pitch circle (N/mm²)	[σHw]	362.37	
Coefficient [fZCa] 1.07 (Helical gear sets with suitable flank modifications acc. to manufacturers experience)			
Single tooth contact factor	[ZB,ZD]	1.03	1.03
Contact stress (N/mm²)	[σHB, σHD]	374.84	374.84
Lubricant coefficient for N_L	[ZL]	0.846	0.846
Speed factor at N_L	[ZV]	1.092	1.092
Roughness factor for N_L	[ZR]	0.940	0.940
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.852	0.954
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.740	0.828
Limited pitting is permissible: Yes			
Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	504.28	515.29
Pitting stress limit (N/mm²)	[σHG]	488.82	499.49

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	0.97	0.97
Safety factor for contact stress on operating pitch circle	[SHw]	1.35	1.38
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.30	1.33
Safety regarding transmittable torque	[SHBD²]	1.70	1.78
Transmittable power (kW)	[kWRating]	540.39	564.24

8 Micropitting

Calculation method according to	ISO/TS 6336-22:2018	
Lubricant load according to FVA Info sheet 54/7 10, TR30 example 7: nu40 input; other oil properties are ancillary defaults		
Reference data FZG-C Test:		
Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	128.014
Contact temperature (°C)	[θBRef,A]	253.676
Lubrication gap thickness (µm)	[hRef,A]	0.083
Specific film thickness in test	[λGFT]	0.167

		------- Gear 1 ---------- Gear 2 -------
Calculation of permissible specific film thickness		
Material coefficient	[WW]	0.500
Permissible specific film thickness	[λGFP]	0.117
Interim results in accordance with ISO/TS 6336-22:2018		
Coefficient of friction	[μ_m]	0.029
Lubricant factor	[XL]	1.000
Roughness factor	[XR]	0.990
Lubrication coefficient for lubrication type	[XS]	1.000
Tooth mass temperature (°C)	[θM]	80.416
Tip relief factor	[XCa(A)]	1.998
Loss factor	[HV]	0.037
Equivalent Young's modulus (N/mm²)	[Er]	226373.626

Pressure-viscosity coefficient (m²/N)	[α38]	0.01589
Dynamic viscosity (Ns/m²)	[ηtM]	17.414
Roughness average value (µm)	[Ra]	1.000

Calculation of speeds, load distribution and flank curvature according to method B following ISO/TS 6336-22:2018.

C_a taken as optimal in the calculation. 0=no, 1=yes		0	0
Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)		0	
Diameter (mm)	[dy]	151.392	769.520
Relative radius of curvature (mm)	[ρred]	21.907	
Load sharing factor	[XY]	0.771	
Contact stress (N/mm²)	[pH]	237.996	
Contact stress (N/mm²)	[pdyn]	424.659	
Minimal specific film thickness	[λGFY]	1.065	(h_Y=1.065 µm)
Safety against micropitting	[Sλ(B)]	9.128	

9 Tooth flank fracture

Failed to calculate the tooth flank fracture. Please check ISO6336-4_res.TMP file for more information.

10 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
Relevant tip relief (µm)	[Ca]	13.00	13.00
Optimal tip relief (µm)	[Ceff]	2.16	
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0
		---------- Pair	----------
Helical load factor for scuffing	[KBγ]	1.300	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	14
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	82.550	
Applicable circumferential force/facewidth (N/mm)	[wBt]	132.756	
$K_{B\gamma}$ = 1.300, $w_{Bt}\cdot K_{B\gamma}$ = 172.582			
Angle factor	[Xαβ]	0.987	
ε_1 =1.029, ε_2 =0.657			

10.1 Flash temperature-criteria

		---------- Pair	----------
Lubricant factor	[XL]	0.867	
Tooth mass temperature (°C)	[θMi]	81.23	
$\theta_{Mi} = \theta_{oil} + X_S\cdot 0.47\cdot X_{mp}\cdot\theta_{flm}$			
Average flash temperature (Δ°C)	[θflm]	2.63	
Scuffing temperature (°C)	[θS]	462.25	
Γ coordinates (point of highest temperature)	[Γ]	0.193	
[Γ.A]= -0.123, [Γ.E]= 0.193			
Maximum flash temperature (Δ°C)	[θflamax]	6.72	
Maximum contact temperature (°C)	[θBmax]	87.96	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Approach factor	[XJ]	1.000	
Load sharing factor	[XΓ]	0.771	
Dynamic viscosity (mPa*s)	[ηM]	17.45	(80.0 °C)
Coefficient of friction	[μ_m]	0.033	

Required safety	[SBmin]	2.000
Margin of safety for scuffing, flash temperature	[SB]	48.047

10.2 Integral temperature-criteria

		Pair
Lubricant factor	[XL]	1.000
Tooth mass temperature (°C)	[θMC]	80.69
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$		
Integral scuffing temperature (°C)	[θSint]	464.55
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058
Running-in factor, well run in	[XE]	1.000
Contact ratio factor	[Xε]	0.228
Dynamic viscosity (mPa*s)	[ηOil]	17.45 (80.0 °C)
Mean coefficient of friction	[μ_m]	0.029
Geometry factor	[XBE]	0.090
Meshing factor	[XQ]	1.000
Tip relief factor	[XCa]	1.998
Mean flash temperature (Δ°C)	[θflaint]	0.98
Integral tooth flank temperature (°C)	[θint]	82.17
Required safety	[SSmin]	1.800
Safety factor for scuffing (intg.-temp.)	[SSint]	5.654
Safety factor for transmitted moment (int.-T.)	[SSL]	177.490

11 Measurements for tooth thickness

11.1 Tooth thickness tolerances

		Gear 1	Gear 2
Tooth thickness tolerance		Own Input	Own Input
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.217 / -0.217	-0.217 / -0.217

11.2 Base tangent lengths

		Gear 1	Gear 2
Number of teeth spanned	[k]	12.000	56.000
Base tangent length (no backlash) (mm)	[Wk]	60.272	289.571
Base tangent length with allowance (mm)	[Wk.e/i]	60.068 /60.068	289.367 /289.367
(mm)	[ΔWk.e/i]	-0.204 / -0.204	-0.204 / -0.204
Diameter of measuring circle (mm)	[dMWk.m]	154.298	766.099

> Gear 2: Base tangent length cannot be measured (gear too thin)

11.3 Measurement over balls and pins

		Gear 1	Gear 2
Theoretical diameter of ball/pin (mm)	[DM]	2.901	2.842
Effective diameter of ball/pin (mm)	[DMeff]	3.000	3.000
Radial single-ball measurement, no backlash (mm)	[MrK]	79.472	385.294
Radial single-ball measurement (mm)	[MrK.e/i]	79.200 /79.200	385.002 /385.002
Diameter of measuring circle (mm)	[dMMr.m]	154.310	765.965
Diametral measurement over two balls, no backlash (mm)	[MdK]	158.943	770.583
Diametral measurement over two balls (mm)	[MdK.e/i]	158.399 /158.399	769.998 /769.998
Diametral measurement over pins, no backlash (mm)	[MdR]	158.943	770.588
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	158.399 /158.399	770.003 /770.003
Measurement over 2 pins, free, AGMA 2002 (mm)	[dk2f.e/i]	0.000 / 0.000	769.997 /769.997
Measurement over 2 pins, transverse, AGMA 2002 (mm)	[dk2t.e/i]	0.000 / 0.000	770.008 /770.008
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	158.399 /158.399	770.003 /770.003

11.4 Tooth thickness

		Gear 1	Gear 2
Medium tip diameter (mm)	[da.m]	157.990	769.520
Reference chordal height from da.m (mm)	[hac]	2.313	2.172
Tooth thickness at height hac, chord (mm)	[sc]	3.105	3.013
(mm)	[sc.e/i]	2.890 / 2.890	2.796 / 2.796
Tooth thickness on reference circle, arc (mm)	[sn]	3.105	3.013
(mm)	[sn.e/i]	2.889 / 2.889	2.796 / 2.796

11.5 Backlash

		Pair
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.000 / 0.000
Radial backlash (mm)	[jrw.e/i]	0.589 / 0.589
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.458 / 0.458
Normal backlash (mm)	[jnw.e/i]	0.407 / 0.407
Center distance allowances (mm)	[Aa.e/i]	0.000 / 0.000
Backlash free center distance (mm)	[aControl.e/i]	459.786 /459.786
Backlash free center distance, allowances (mm)	[jta]	-0.589 / -0.589

		Gear 1	Gear 2
dNf.i with aControl (mm)	[dNf0.i]	150.370	762.008
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	-0.218	-0.100
Tip clearance (mm)	[c0.i(aControl)]	0.131	0.132

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.0684/ 0.0684	(0°4'6"/ 0°4'6")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.3412/ 0.3412	(0°20'28"/ 0°20'28")

12 Gear tolerances

		Gear 1		Gear 2	
According to ISO 1328-1:2013, ISO 1328-2:1997		[A]		[A]	
Single pitch tolerance (µm)	[fpT]	± 6.00	(-)	± 6.50	(-)
Base pitch tolerance (µm)	[fpbT]	± 5.60	(-)	± 6.07	(-)
Sector pitch tolerance (µm)	[Fpz/8T]	± 13.00	(5)	± 18.00	(5)
Cumulative pitch tolerance, total (µm)	[FpT]	20.00	(5)	30.00	(5)
Adjacent pitch difference tolerance (µm)	[fuT]	8.00	(5)	9.00	(5)
Profile form tolerance (µm)	[ffαT]	6.00	(-)	6.00	(-)
Profile slope tolerance (µm)	[fHαT]	± 4.80	(5)	± 5.50	(5)
Profile tolerance, total (µm)	[FαT]	7.50	(5)	8.00	(5)
Helix form tolerance (µm)	[ffβT]	9.00	(5)	10.00	(5)
Helix slope tolerance (µm)	[fHβT]	± 8.00	(-)	± 8.50	(-)
Helix tolerance, total (µm)	[FβT]	12.00	(5)	13.00	(5)
Runout tolerance (µm)	[FrT]	18.00	(5)	27.00	(5)
Single flank composite tolerance, total (µm)	[FisT]	26.00	(5)	36.00	(5)
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	5.50	(5)	5.50	(5)
Radial composite tolerance, total (µm)	[Fi"]	26.00	(5)	40.00	(5)
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	6.50	(5)	7.00	(5)

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

		Gear 1		Gear 2	
According to ISO 1328-2:2020		[R]		[R]	
Radial composite tolerance, total (µm)	[FidT]	23.00	(37)	39.00	(39)
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	9.50	(37)	16.00	(39)

Own input for tolerance: (-)

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99

Total helix deviation for calculation (µm)	[Fβ]	13.00
Distance between bearings for calculation (mm)	[LG]	355.60
Maximum value for axis deviation error (µm)	[fΣβ]	± 10.71
Maximum value for axis inclination error (µm)	[fΣδ]	± 21.41

13 Modifications and determination of the tooth form

13.1 Profile and flank line modifications

Gear 1

Symmetric (both flanks)

- Tip relief, linear

Caa = 13.000 µm LCa = 0.500 ·mn dCa = 157.283 mm

Gear 2

Symmetric (both flanks)

- Tip relief, linear

Caa = 13.000 µm LCa = 0.500 ·mn dCa = 768.893 mm

13.1.1 Tip relief verification

		------- Gear 1 ----------	Gear 2 -------
Diameter (mm)	[dcheck]	157.956	769.486
Tip relief left/right (µm)	[Ca L/R]	12.4 /12.4	12.3 /12.3
Diameter (mm)	[dNa.i]	157.990	769.520
Tip relief left/right (µm)	[Ca L/R]	13.0 /13.0	13.0 /13.0

13.2 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

14 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	461.692

14.1 Masses, stiffnesses and moments of inertia

		------- Gear 1 ----------	Gear 2 -------
Mass (kg)	[m]	31.565	778.471
Total mass (kg)	[Σm]	810.036	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.09380	57.05462
System (da+df)/2...di (kg*m²)	[J]	2.38664	

14.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	13.840
Torsion when subjected to nominal torque (°)	[δcr]	0.002
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	99.811

14.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.028
Wear sliding coefficient by Niemann	[ζw]	0.294

Loss factor	[HV]	0.037
Gear power loss (kW)	[PVZ]	0.307
Meshing efficiency (%)	[ηz]	99.897

14.4 Indications for the manufacturing by wire cutting

		------- Gear 1	---------- Gear 2 -------
Deviation from theoretical tooth trace (µm)	[WireErr]	1213.0	243.8
Permissible deviation (µm)	[Fb/2]	6.0	6.5

15 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.3387
Required safety for tooth flank	[S_{Hmin}]	0.9693
Required service life	[H]	30000.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	>	1000000

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	1e+06	1e+06

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

15.1 Damage

Damage relative to the required service life (H, 30000.0 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	0.0000	0.0000	0.0000

16 Reliability calculation

16.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

16.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	<TD CENTER>β	<TD CENTER>t_0	<TD CENTER>T	<TD CENTER>R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	324000	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	324000	1.300e+00	9.014e+29	4.295e+30	100.0000
2	Tooth root	64951	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	64951	1.300e+00	9.014e+29	4.295e+30	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

16.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	30000.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000

Reliability R, tooth flanks subsystem (%)	[R_{subH}]	100.0000
Reliability R, gears subsystem (%)	[R_{subG}]	100.0000
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	> 1'000'000
Service life H, gears subsystem (h)	[H_{subG}]	> 1'000'000

17 Remarks

17.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

17.2 Calculations and factors

- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with Z_β = 1/(COS(β)^0.5)
- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 707)
